THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: FF-W-84

Form Approved
OMB No. 0704-0188

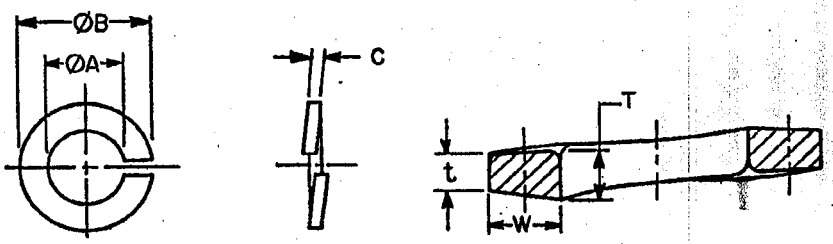



TABLE I. DASH NUMBERS AND DIMENSIONS

| NOMINAL SIZE | CARBON STEEL, PLAIN 2/ | CARBON STEEL, CAD OR ZN 2/ | CARBON STEEL, CAD | CARBON STEEL, PHOS 2/ | CRES 302 OR 420 2/ | PH BRZ OR TIN BRS | NICKEL COPPER ALLOY | CRES 300 SERIES | CRES 420 | ØA (ID) MIN | ØA (ID) MAX | ØB (CD) MAX | WASHER SECTION W WIDTH MIN | WASHER SECTION C THICK $\frac{T+t}{2}$ MIN 1/ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | DASH NUMBERS | | | | | | | | | | | | | |
| NO.2 (.086) | 1 | 20 | 39 | 58 | 77 | 96 | 115 | 134 | 153 | .088 | .094 | .172 | .035 | .020 |
| NO.4 (.112) | 2 | 21 | 40 | 59 | 78 | 97 | 116 | 135 | 154 | .114 | .120 | .209 | .040 | .025 |
| NO.6 (.138) | 3 | 22 | 41 | 60 | 79 | 98 | 117 | 136 | 155 | .141 | .148 | .250 | .047 | .031 |
| NO.8 (.164) | 4 | 23 | 42 | 61 | 80 | 99 | 118 | 137 | 156 | .167 | .174 | .293 | .055 | .040 |
| NO.10 (.190) | 5 | 24 | 43 | 62 | 81 | 100 | 119 | 138 | 157 | .193 | .200 | .334 | .062 | .047 |
| 1/4 (.250) | 6 | 25 | 44 | 63 | 82 | 101 | 120 | 139 | 158 | .252 | .260 | .487 | .109 | .062 |
| 5/16 (.312) | 7 | 26 | 45 | 64 | 83 | 102 | 121 | 140 | 159 | .314 | .322 | .583 | .125 | .078 |
| 3/8 (.375) | 8 | 27 | 46 | 65 | 84 | 103 | 122 | 141 | 160 | .377 | .385 | .680 | .141 | .094 |
| 7/16 (.437) | 9 | 28 | 47 | 66 | 85 | 104 | 123 | 142 | 161 | .440 | .450 | .776 | .156 | .109 |
| 1/2 (.500) | 10 | 29 | 48 | 67 | 86 | 105 | 124 | 143 | 162 | .502 | .512 | .869 | .171 | .125 |
| 9/16 (.562) | 11 | 30 | 49 | 68 | 87 | 106 | 125 | 144 | 163 | .564 | .574 | .965 | .188 | .141 |
| 5/8 (.625) | 12 | 31 | 50 | 69 | 88 | 107 | 126 | 145 | 164 | .628 | .640 | 1.072 | .203 | .156 |
| 3/4 (.750) | 13 | 32 | 51 | 70 | 89 | 108 | 127 | 146 | 165 | .753 | .765 | 1.264 | .234 | .188 |
| 7/8 (.875) | 14 | 33 | 52 | 71 | 90 | 109 | 128 | 147 | 166 | .878 | .890 | 1.455 | .266 | .219 |
| 1 (1.000) | 15 | 34 | 53 | 72 | 91 | 110 | 129 | 148 | 167 | 1.003 | 1.015 | 1.647 | .297 | .250 |
| 1-1/8 (1.125) | 16 | 35 | 54 | 73 2/ | 92 | 111 | 130 | 149 | 168 | 1.129 | 1.144 | 1.838 | .328 | .281 |
| 1-1/4 (1.250) | 17 | 36 | 55 | 74 2/ | 93 | 112 | 131 | 150 | 169 | 1.254 | 1.272 | 2.028 | .359 | .312 |
| 1-3/8 (1.375) | 18 | 37 | 56 | 75 2/ | 94 | 113 | 132 | 151 | 170 | 1.379 | 1.399 | 2.210 | .391 | .344 |
| 1-1/2 (1.500) | 19 | 38 | 57 | 76 2/ | 95 | 114 | 133 | 152 | 171 | 1.504 | 1.524 | 2.409 | .422 | .375 |

1/ SEE REQUIREMENT 4

2/ SEE NOTE 2

(J) ENTIRE STANDARD REVISED.

INCH - POUND

PREPARING ACTIVITY: ARMY-AR
CUSTODIANS: ARMY- AR NAVY- AS
AIR FORCE- 99 DLA-
REVIEW: AT,AV,GL,MI,IS,NS
USER: ER,ME,MC
PROJECT NUMBER: 5310-1609

MILITARY SPECIFICATION SHEET

TITLE
WASHER, LOCK - SPRING, HELICAL, REGULAR (MEDIUM) SERIES

SPECIFICATION SHEET NUMBER
MS35338J 30 MARCH 1990

SUPERSEDING
MS35338H 29 NOV 1983 (SEE NOTE 3)

AMSC- N/A FSC - 5310

DISTRIBUTION STATEMENT
A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T17:21Z
Check the source to verify that this is the current version before use.



DD Form 672, MAY 88 PREVIOUS EDITIONS ARE OBSOLETE

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: FF-W-84

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

Form Approved
OMB No. 0704-0188

REQUIREMENTS:

1. MATERIAL:

   a. STEEL, CARBON, GRADE 1060 (UNS G10600) THRU 1080 (UNS G10800) IN ACCORDANCE WITH ASTM A29.

   b. CORROSION-RESISTANT STEEL (NON-MAGNETIC) WASHERS SHALL BE TYPE 316 (UNS S31600) THRU 1/4 (.250) AND TYPE 300 SERIES (UNS S30200 TO UNS S31600) FOR ALL OTHER SIZES, AND CORROSION-RESISTANT STEEL (MAGNETIC) WASHERS SHALL BE TYPE 420 (UNS S42000).

   c. PHOSPHOR BRONZE, COPPER ALLOY NO. 510 (UNS C51000) IN ACCORDANCE WITH ASTM B159.

   d. TIN-BRASS, COMPOSITION 425 (UNS C42500) IN ACCORDANCE WITH ASTM B591.

   e. NICKEL-COPPER-ALUMINUM ALLOY, FORM 2 (UNS N05500) IN ACCORDANCE WITH QQ-N-286.

2. PROTECTIVE FINISHES:

   a. CARBON STEEL WASHERS:

      1. PHOSPHATE COATING SHALL BE IN ACCORDANCE WITH MIL-P-16232, TYPE Z, CLASS 2.

      2. CADMIUM PLATING SHALL BE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2, OR MIL-C-81562, TYPE II, CLASS 2 (NO SUBSTITUTES ALLOWED).

   b. CORROSION-RESISTANT STEEL WASHERS:

      1. PASSIVATION SHALL BE IN ACCORDANCE WITH QQ-P-35.

      2. BLACK OXIDE COATING (FOR CRES 300 SERIES ONLY) SHALL BE IN ACCORDANCE WITH MIL-C-13924, CLASS 4. (SEE REQUIREMENT 6).

   c. PHOSPHOR BRONZE AND TIN-BRASS WASHERS SHALL BE CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3 AND SHALL BE DYED PREDOMINANTLY YELLOW TO MATCH COLOR 13415 OR 13432 OR 13507 OR 23538 IN ACCORDANCE WITH FED-STD-595.

   d. NICKEL-COPPER-ALUMINUM ALLOY WASHERS - UNCOATED.

3. HARDNESS:

   a. CARBON STEEL WASHERS SHALL HAVE A HARDNESS OF 38 TO 46HRC.

   b. CORROSION-RESISTANT STEEL WASHERS TYPE 420 (UNS S42000) SHALL HAVE A HARDNESS OF 45-53HRC, AND TYPE 300 (UNS S3XXXX) SERIES WASHERS SHALL HAVE A HARDNESS OF 35-43HRC.

   c. PHOSPHOR BRONZE AND TIN-BRASS WASHERS SHALL HAVE A MINIMUM HARDNESS OF 90HRB.

   d. NICKEL-COPPER-ALUMINUM ALLOY WASHERS SHALL HAVE A HARDNESS OF 33-40HRC.

4. MAXIMUM TOLERANCE FOR NOMINAL SERIES .086 THRU .190 SHALL BE +.006, .250 THRU .750 SHALL BE +.010, AND .875 THRU 1.500 SHALL BE .020 (SEE TABLE I).

5. MAGNETIC PERMEABILITY: CORROSION-RESISTANT STEEL WASHERS (300 SERIES ONLY) SHALL HAVE A MAGNETIC PERMEABILITY OF 2.0 MAXIMUM (AIR = 1.0) FOR FIELD STRENGTH OF H = 200 OERSTEDS USING A MAGNETIC INDICATOR IN ACCORDANCE WITH MIL-I-17214.

6. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC SPECIFICATION SHEET NUMBER FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE 1. WHEN BLACK OXIDE COATING IS REQUIRED, THE DASH NUMBER SHALL BE FOLLOWED BY THE LETTER "B".

EXAMPLE: MS35338-48

- 48 — DASH NUMBER
- MS35338 — BASIC SPECIFICATION SHEET NUMBER

MS35338-48 INDICATES - WASHER, LOCK-SPRING, HELICAL-SPRING, REGULAR (MEDIUM) SERIES, 1/2 (.500) NOMINAL DIAMETER, CARBON STEEL; CADMIUM PLATED

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. CANCELLED, IN PART, AND INACTIVE FOR DESIGN. FOR SUBSTITUTIONS SEE TABLE II.
3. MS35338 SUPERSEDES MS16214, MS35337, AN935, NAS1061, BECK1 AND BECK2, SEE TABLE II.
4. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
5. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DoDISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

| | | |
|---|---|---|
| PREPARING ACTIVITY: ARMY-AR<br>CUSTODIANS: ARMY-AR NAVY-AS<br>AIR FORCE-99 DLA-<br>REVIEW: AT,AV,GL,MI,IS,NS<br>USER: ER,ME,MC<br>PROJECT NUMBER: 5310-1609 | MILITARY SPECIFICATION SHEET<br>TITLE<br>WASHER, LOCK - SPRING, HELICAL, REGULAR (MEDIUM) SERIES | SPECIFICATION SHEET NUMBER<br>**MS35338J** 30 MARCH 1990<br>SUPERSEDING<br>MS35338H 29 NOV 1983 (SEE NOTE 3)<br>AMSC- N/A FSC- 5310 |

DISTRIBUTION STATEMENT

A. Approved for public release; distribution is unlimited.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T17:21Z
Check the source to verify that this is the current version before use.

DD Form 672, MAY 88 PREVIOUS EDITIONS ARE OBSOLETE

GPO: 1989-606-045

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: FF-W-84

Form Approved
OMB No. 0704-0188

TABLE II. INTERCHANGEABILITY.

After 21 March 1966, (1) washers given on MS16214, MS35337, NAS1061, and this standard in part, are inactive for design and are cancelled, but their existing stocks should be used for maintenance purposes until depleted and (2) washers given on AN935 are inactivated for new design. Use only the following substitutive washers for new design and replacement.

| CROSS REFERENCE OF PART NUMBERS | | | | | | | |
|---|---|---|---|---|---|---|---|
| CANCELLED INACTIVATED | SUBSTITUTIVE | CANCELLED INACTIVATED | SUBSTITUTIVE | CANCELLED INACTIVATED | SUBSTITUTIVE | CANCELLED INACTIVATED | SUBSTITUTIVE |
| MS16214-1 | MS35338-135 | MS35337-37 | MS35338-56 | MS35337-108 | MS35338-108 | AN935-2 | MS35338-36 |
| 2 | 136 | 38 | 57 | 109 | 109 | -3 | 1/ |
| 3 | 137 | 39 | 58 | 110 | 110 | -4 | 40 |
| 4 | 138 | 40 | 40 | 111 | 111 | -5 | 1/ |
| 5 | 139 | 41 | 41 | 112 | 112 | -6 | 41 |
| 6 | 140 | 42 | 42 | 113 | 113 | -8 | 42 |
| 7 | 141 | 43 | 43 | 114 | 114 | -10 | 43 |
| 8 | 142 | 44 | 44 | | | -12 | 1/ |
| 9 | 143 | 45 | 45 | MS35338-1 | MS35338-39 | -416 | 44 |
| 10 | 145 | 46 | 46 | 2 | 40 | -516 | 45 |
| 11 | 146 | 47 | 47 | 3 | 41 | -616 | 46 |
| 12 | 147 | 48 | 48 | 4 | 42 | -716 | 47 |
| 13 | 148 | 49 | 49 | 5 | 43 | -816 | 48 |
| 14 | 149 | 50 | 50 | 6 | 44 | -916 | 49 |
| 15 | 150 | 51 | 51 | 7 | 45 | -1016 | 50 |
| 16 | 151 | 52 | 52 | 8 | 46 | -1216 | 51 |
| 17 | 152 | 53 | 53 | 9 | 47 | -2L | 39 |
| 18 | 97 | 54 | 54 | 10 | 48 | -3L | 1/ |
| 19 | 98 | 55 | 55 | 11 | 49 | -4L | 40 |
| 20 | 99 | 56 | 56 | 12 | 50 | -5L | 1/ |
| 21 | 100 | 57 | 57 | 13 | 51 | -6L | 41 |
| 22 | 101 | 58 | 58 | 14 | 52 | -8L | 42 |
| 23 | 102 | 59 | 59 | 15 | 53 | -10L | 43 |
| 24 | 103 | 60 | 60 | 16 | 54 | -12L | 1/ |
| 25 | 104 | 61 | 61 | 17 | 55 | -416L | 44 |
| 26 | 105 | 62 | 62 | 18 | 56 | -516L | 45 |
| 27 | 107 | 63 | 63 | 19 | 57 | -616L | 46 |
| 28 | 108 | 64 | 64 | 20 | 39 | -716L | 47 |
| 29 | 109 | 65 | 65 | 21 | 40 | -816L | 48 |
| 30 | 110 | 66 | 66 | 22 | 41 | -916L | 49 |
| 31 | 111 | 67 | 67 | 23 | 42 | -1016L | 50 |
| 32 | 112 | 68 | 68 | 24 | 43 | -1216L | 51 |
| 33 | 113 | 69 | 69 | 25 | 44 | B2 | 96 |
| 34 | 114 | 70 | 70 | 26 | 45 | B3 | 1/ |
| | | 71 | 71 | 27 | 46 | B4 | 97 |
| MS35337-1 | MS35338-39 | 72 | 72 | 28 | 47 | B5 | 1/ |
| 2 | 40 | 73 | 58 | 29 | 48 | B6 | 98 |
| 3 | 41 | 74 | 55 | 30 | 49 | B8 | 99 |
| 4 | 42 | 75 | 56 | 31 | 50 | B10 | 100 |
| 5 | 43 | 76 | 57 | 32 | 51 | B12 | 1/ |
| 6 | 44 | 77 | 134 OR 153 | 33 | 52 | B416 | 101 |
| 7 | 45 | 78 | 135 OR 154 | 34 | 53 | B516 | 102 |
| 8 | 46 | 79 | 136 OR 155 | 35 | 54 | B616 | 103 |
| 9 | 47 | 80 | 137 OR 156 | 36 | 55 | B716 | 104 |
| 10 | 48 | 81 | 138 OR 157 | 37 | 56 | B816 | 105 |
| 11 | 49 | 82 | 139 OR 158 | 38 | 57 | B916 | 106 |
| 12 | 50 | 83 | 140 OR 159 | 73 | 54 | B1016 | 107 |
| 13 | 51 | 84 | 141 OR 160 | 74 | 55 | B1216 | 108 |
| 14 | 52 | 85 | 142 OR 161 | 75 | 56 | B2L | 96 |
| 15 | 53 | 86 | 143 OR 162 | 76 | 57 | B3L | 1/ |
| 16 | 54 | 87 | 144 OR 163 | 77 | 134 OR 153 | B4L | 97 |
| 17 | 55 | 88 | 145 OR 164 | 78 | 135 OR 154 | B5L | 1/ |
| 18 | 56 | 89 | 146 OR 165 | 79 | 136 OR 155 | B6L | 98 |
| 19 | 57 | 90 | 147 OR 166 | 80 | 137 OR 156 | B8L | 99 |
| 20 | 39 | 91 | 148 OR 167 | 81 | 138 OR 157 | B10L | 100 |
| 21 | 40 | 92 | 149 OR 168 | 82 | 139 OR 158 | B12L | 1/ |
| 22 | 41 | 93 | 150 OR 169 | 83 | 140 OR 159 | B416L | 101 |
| 23 | 42 | 94 | 151 OR 170 | 84 | 141 OR 160 | B516L | 102 |
| 24 | 43 | 95 | 152 OR 171 | 85 | 142 OR 161 | B616L | 103 |
| 25 | 44 | 96 | | 86 | 143 OR 162 | B716L | 104 |
| 26 | 45 | 97 | 97 | 87 | 144 OR 163 | B816L | 105 |
| 27 | 46 | 98 | 98 | 88 | 145 OR 164 | B916L | 106 |
| 28 | 47 | 99 | 99 | 89 | 146 OR 165 | B1016L | 107 |
| 29 | 48 | 100 | 100 | 90 | 147 OR 166 | B1216L | 108 |
| 30 | 49 | 101 | 101 | 91 | 148 OR 167 | | |
| 31 | 50 | 102 | 102 | 92 | 149 OR 168 | NAS1061-6 | MS35338-136 |
| 32 | 51 | 103 | 103 | 93 | 150 OR 169 | 10 | 138 |
| 33 | 52 | 104 | 104 | 94 | 151 OR 170 | 416 | 139 |
| 34 | 53 | 105 | 105 | 95 | 152 OR 171 | 516 | 140 |
| 35 | 54 | 106 | 106 | | | 616 | 141 |
| 36 | 55 | 107 | 107 | | | | |

1/ No replacement

PREPARING ACTIVITY: ARMY-AR
CUSTODIANS: ARMY- AR NAVY- AS
AIR FORCE- 99 DLA-
REVIEW: AT,AV,GL,MI,SS,NS
USER: ER,ME,MC
PROJECT NUMBER: 5310-1609

MILITARY SPECIFICATION SHEET

TITLE
WASHER, LOCK - SPRING, HELICAL, REGULAR (MEDIUM) SERIES

SPECIFICATION SHEET NUMBER
MS35338J 30 MARCH 1990

SUPERSEDING
MS35338H 29 NOV 1983 (SEE NOTE 3)

AMSC- N/A FSC - 5310

DISTRIBUTION STATEMENT
A. Approved for public release; distribution is unlimited.

Downloaded from http://assist.daps.mil -- Downloaded: 2026-10-23T17:21Z
Check the source to verify that this is the current version before use.



DD Form 672, MAY 88 PREVIOUS EDITIONS ARE OBSOLETE

GPO. 1988-606-048